

- MN sends binding update to home agent.
- Home agent uses proxy neighbour discovery to record MN in home network (HN).
- IP datagram addressed to MN is encapsulated in IPv6 to IPv6 tunnel. Then, the encapsulated packet is sent to COA of MN.
- IP datagram from MN makes use of the same tunnel in reverse.
- IP datagram from MN is sent to correspondent node directly.
- MN sends binding update to correspondent node.
- Mobility support in IPv6 retains the ideas of HN, home agent and use of encapsulation to deliver packets from HN to MN's current point of attachment, like in IPv4.
- Foreign agents are not required to support mobility in IPv6. IPv6 improves routing efficiency, enhances mobility support and simplifies procedures.

IP telephony and VoIP

The Internet was established primarily for transporting traditional data. Traditional data differs from other communication services, namely, voice and video, in many respects. From customers' point of view, a single network for all services (voice, video and data) sounds appealing.

Single infrastructure, single access is always affordable and convenient. The natural question that arose then was, "Why not use the Internet to carry all services of voice, video and data?"

Solutions for this came in the forms of Internet Protocol telephony (IP telephony)/Voice over Internet Protocol (VoIP) and Internet Protocol TV (IPTV). These solutions address the technical challenges of carrying real-time and jitter-free voice and video over the Internet.

There are several advantages for transmitting VoIP, including:

1. Long-distance calls at low cost but,

maybe, of low quality

2. Cheaper two-in-one service
3. Use of PC as a true multimedia terminal
4. One connection for all services
5. Local exchanges to support telephone with the Internet as backbone and without high investment in expensive backbone infrastructure
6. Use of packetised voice allows voice compression, which, in turn, decreases transmission time and cost. Earlier, telecommunication traffic or telephony connections outnumbered data traffic.

In the future, there will be explosion in data traffic. The need to deploy the Internet for real-time services like voice and video has necessitated redesigning of some features of the Internet. The two important features related to this emerging issue are:

- Redesigning of IP datagram format
- Using real-time data transfer protocol (RTP) and IP for carrying voice over conventional IP datagram and the Internet

Internet telephony refers to communications services—voice, facsimile and/or voice-messaging applications—that are transported via the Internet, rather than public switched telephone network (PSTN).

IP telephony is a general term used for technologies that use IP's packet switched connections to exchange voice, fax and other forms of information that have traditionally been carried over dedicated circuit-switched connections of PSTN.

Using the Internet, calls travel as packets of data on shared lines, avoiding tolls of PSTN. The challenge in IP telephony is to deliver voice, fax or video packets in a dependable flow to the user. Much of IP telephony focuses on this challenge. VoIP is an organised effort to standardise IP telephony.

IP telephony is an important part of the integration of computers, telephones and TVs in a single, integrated information environment. It has become a reliable communication option with a well-established standardised communication protocol.

VoIP is simply transmission of voice traffic over IP-based networks. PSTN based on circuit switching provides voice service with guaranteed quality of service (QoS). This is not the case with voice service provided by the Internet that acts on packet switching.

There are many technical challenges that a voice packet faces while in transition over packet switching network like the Internet. These include packet loss, packet transfer delay and jittering delay.

Voice communication involves human interaction. As such, loss of some voice packets can be tolerated due to human intelligence and perception involved in recovery. However, excessive loss of voice packets may seriously degrade voice quality. Moreover, PSTN is a reliable voice service provider, whereas the Internet is not, because it is datagram-based.

Delay is a more serious issue for real-time interactive services like voice. It is the time difference between the time the sender releases the packet to the network and the time at which the receiver receives the packet from the network. Delay refers to:

- Total transfer delay of a packet that includes coding/decoding delay, propagation delay, transmission delay, node processing and queue delay, switching and routing delay
- Jittering delay that refers to phase delay between two successive packets

If total delay exceeds a certain value, users may get irritated with the service. According to a statistic, a delay up to eighty milli-seconds